# MUNSHI SAIFUZZAMAN

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## EDUCATION

**Utah State University** — Logan, UT

- *Doctor of Philosophy (Ph.D.) in Computer Science* — *Aug 24 – May 29 (Expected)*
  - CGPA: **4.0 (A)** on a scale of 4.00
  - **Research Focus:** Security, privacy, and resilience of mobile networks, and AI-enabled systems. (**Refer to [C1], [I1], and [J1] publications)**
- *Master of Computer Science* — *Aug 24 – Dec 26 (Expected)*
  - **Courses Taken:** Advance Algorithms, Advance Database, Program Analysis and Its Applications, Computer Network & Security, Usable Privacy & Security, Human Factors in Computing.

**Shahjalal University of Science and Technology** — Sylhet, BD

*BSc. (Engg.) in Computer Science and Engineering* — *Feb 16 – Jun 21*

- CGPA: **3.79 (A)** on a scale of 4.00
- **6th** in class of **107 students** (Acknowledgement)
- **Thesis:** Assessing and developing differential privacy based algorithms for privacy-preserving wearable data publishing. (**Refer to [J4] and [J2] publications)**

## APPOINTMENT

**Utah State University** — Logan, UT

- *Graduate Instructor* (CS 1030) — *Jan 27 – May 27 (5 months)*
- *Graduate Teaching Assistant* (CS 2610) — *Aug 26 – Dec 26 (5 months)*
- *Building Representative* — *May 26 – Aug 26 (4 months)*
- *Graduate Teaching Assistant* (CS 2610) — *Aug 25 – Apr 26 (9 months)*

**Utah State University** — Logan, UT

*Graduate Teaching Assistant* (CS 2610) — *Aug 24 – May 25 (9 months)*

**Dynamic Solution innovators Limited** — Dhaka, BD

- *Assistant Software Engineer* (Exp letter) — *May 23 – Jul 24 (13 months)*
- *Junior Software Engineer* — *May 21 – May 23 (26 months)*

**Daffodil International University** — Dhaka, BD

*Adjunct Lecturer* — *Jul 22 – Dec 22 (6 months)*

**Shahjalal University of Science & Technology** — Dhaka, BD

*Undergraduate Research Assistant* — *Jul 20 – Nov 20 (5 months)*

**WIB Corporation** — Remote

*Software Engineer* (Exp letter) — *Jul 20 – Aug 20 (2 months)*

**Shahjalal University of Science & Technology** — Dhaka, BD

*Undergraduate Research Assistant* — *Aug 18 – Dec 18 (5 months)*

## RESEARCH INTERESTS

♣ My research focuses on securing next-generation (4G/5G and beyond) *mobile networks*, with emphasis on *privacy-preserving architectures* and identifier protection. I also explore *differential privacy techniques* for real-time *medical and wearable data sharing*, aiming to balance strong privacy guarantees with practical data utility.

## RESEARCH EXPERIENCE

♣ **End-to-End Security Modeling for Wearable Systems** *Apr 26 - May 26*
Developed a Petri net-based modeling framework to represent end-to-end wearable health data pipelines and systematically analyze disruptions caused by both system failures and adversarial attacks. Demonstrated that different disruption sources can produce indistinguishable observable outcomes, providing a foundation for system-level security analysis and resilient wearable system design. (**Refer to [W1] publication)**

♣ **Risk Communication for Password Breach Alerts** *Jan 26 - May 26*
Designed and evaluated a multi-step, contextualized password breach notifier to improve users' understanding of security risks and encourage secure decision-making. Conducted a qualitative user study with 12 participants, showing that reflective and context-aware warning designs improve risk communication and user intention compared to existing browser alerts. (**Refer to [P1] publication)**

♣ **The End-to-end Security in 5G/4G Cellular Networks: A Survey** *Aug 24 - Jan 26*
Led a comprehensive survey of 153 peer-reviewed studies to classify and analyze security threats, mitigation techniques, and open challenges across 4G/5G cellular networks, providing a unified end-to-end taxonomy across mobile network layers. (**Refer to [J1] publication)**

♣ **Privacy-Preserving Data Publishing for Wearable Health Devices** *Feb 20 - Sep 25*
Conducted a systematic literature review on differential privacy in wearables to identify technical gaps, propose adaptive budgeting strategies, and reduce error rates. Based on the findings, we later designed and evaluated ResCoDP, a novel privacy-preserving mechanism for wearable health data publishing. (**Refer to [J4] and [J2] publications)**

♣ **Dissecting Privacy-Exposing Identifiers in 5G/4G Networks** *May 25 - Aug 25*
Conducted the first systematic, standards-based analysis of underexplored privacy-exposing identifiers in 4G/5G cellular networks, revealing critical cross-layer vulnerabilities through real-world validation on a U.S. mobile network. (**Refer to [C1] publication)**

♣ **Assessing Carbon Emission Quantification of Machine Learning** *Feb 25 - Jul 25*
Conducted a comprehensive review of over 200 peer-reviewed studies to analyze carbon emission sources, quantification methods, sustainability challenges, and mitigation strategies in machine learning. Proposed a practical framework and future research roadmap for developing environmentally sustainable machine learning systems. (**Refer to [J3] publication)**

♣ **The Role of the IoT and Machine Learning in Smart Healthcare** *Feb 24 - Jul 24*
Authored a comprehensive overview of machine learning, covering its core concepts, historical evolution, paradigms, challenges, and real-world applications across domains such as healthcare, business, and personalized systems. (**Refer to [B1] publication)**

♣ **IoT and ML for Information Management: A Smart Healthcare Perspective** *Jul 23 - Dec 23*

Conducted a study on IoT-based healthcare systems, analyzing architectural, security, and trust challenges, and explored how machine learning can support data-driven decision-making in smart health environments. (**Refer to [B2] and [B3] publications)**

♣ **Assessing Blockchain Technology to Manage Health Insurance** *Jul 21 - Dec 21*
Explored key challenges in the health insurance industry and proposed blockchain-based solutions to enhance transparency, reduce fraud, and improve data privacy and trustworthiness in healthcare delivery. (**Refer to [B4] publication)**

♣ **Extensive analysis on Question Answering Framework** | *Python* Spring 20
Evaluated three transformer-based models (BERT, DistilBERT, and SQuAD variants) for performance in question answering tasks. [Project Link]

♣ **Wine Quality Test** | *Python* Spring 20
Developed an interface to predict red wine quality using a data mining-based machine learning approach. [Project Link, Demonstration Link]

## HONORS AND AWARDS

♣ Tuition Waiver 2024, 2025
Awarded scholarship from *School of Computing* for consecutive second years during my PhD journey.
♣ Best Young Entrepreneurs 2018
- ♠ Awarded in *Digital Innovation Fair* arranged by *Divisional Commissioner's Office, Sylhet*. The idea was to ensure delivering government rations to the correct people in a corruption-free manner. [Project Link, Demonstration Link]

♣ Government Student Scholarship 2009, 2014
Awarded by the Government of Bangladesh for outstanding academic performance in Grade 5 and Secondary School Examinations.

## ACADEMIC SERVICE

♣ **Conference Reviewer**
- ♠ ACM/IEEE Conference on Connected Health: Applications, Systems and Engineering Technologies (six papers)
- ♠ IEEE Conference on Dependable and Secure Computing 2025 (four papers)

♣ **Journal Co-reviewer** (Requested to assist in review)
- ♠ Free5GC World Forum 2025 (one paper)
- ♠ IEEE Transactions on Reliability (one paper)
- ♠ IEEE Transactions on Vehicular Technology (one paper)
- ♠ Springer Nature Cybersecurity (one paper)

## PEER-REVIEWED PUBLICATIONS

♣ **In-progress**

I1. Ke Xie, Xingyi Zhao, Yiwen Hu, **Munshi Saifuzzaman**, Wen Li, Shuhan Yuan, Tian Xie, Guan-Hua Tu. "*CellSecInspector: Safeguarding Cellular Networks via Automated Security Analysis on Specifications.*" *ACM Annual International Conference on Mobile Computing and Networking (2025)*. **[Submitted]**

♣ **Workshops**

W1. **Munshi Saifuzzaman**, and Tian Xie. "*Attack and Failure Indistinguishability in Wearable Systems: An End to End Pipeline Modeling Approach.*" Proceedings of the

ACM/IEEE International Conference on Connected Health: Applications, Systems and Engineering Technologies *(2026)*.

### ♣ Poster Papers

P1. **Munshi Saifuzzaman**, Weston Fausett, Rebecca Lamoreaux, Abraham Done, Rizu Paudel, and Mahdi Nasrullah Al-Ameen. "*Risk Communication to End Users: Exploring Design Opportunities with Password Breach Alert.*" Companion Publication of the 2026 Conference on Computer-Supported Cooperative Work and Social Computing *(2026)*. **[Acceptance rate: 43.98%]**

### ♣ Conferences

C1. **Munshi Saifuzzaman**, Ke Xie, Tian Xie, Xiao Zhang, and Xinyu Lei. "*Dissecting Privacy-Exposing Identifiers in 5G/4G Networks.*" IEEE Conference on Dependable and Secure Computing *(2025)*. **[Best paper award, Acceptance rate: 40.9%]**

### ♣ Journals [Total IF: 9.9]

J1. **Saifuzzaman, Munshi**, Ke Xie, Xiao Zhang, Yiwen Hu, Xinyu Lei, Guan-Hua Tu, Sihan Wang, Wen Li, and Tian Xie. "*The End-to-end Security in 5G/4G Cellular Networks: A Survey.*" *ACM Computing Survey (2025)*. **[Submitted]**

J2. **Saifuzzaman, Munshi**, Tajkia Nuri Ananna, Mohammad Jabed Morshed Chowdhury, Muneeb-Ul-Hassan, Md Sadek Ferdous, and Farida Chowdhury. "*ResCoDP: A Differentially Private Mechanism for Preserving Privacy in Wearable Physiological Data.*" Springer Computing (2025). **[Impact Factor: 2.8]**

J3. Hasan, Syed Mhamudul, Taminul Islam, **Munshi Saifuzzaman**, Khaled R. Ahmed, Chun-Hsi Huang, and Abdur R. Shahid. "*Carbon Emission Quantification of Machine Learning: A Review.*" *IEEE Transactions on Sustainable Computing (2025)*. **[Impact Factor: 3.9]**

J4. **Saifuzzaman, Munshi**, Tajkia Nuri Ananna, Mohammad Jabed Morshed Chowdhury, Md Sadek Ferdous, and Farida Chowdhury. "*A systematic literature review on wearable health data publishing under differential privacy.*" International Journal of Information Security 21, no. 4 (2022): 847-872. **[Impact Factor: 3.2]**

### ♣ Book Chapters

B1. **Saifuzzaman, Munshi**, and Tajkia Nuri Ananna. '*Introduction to machine learning.*' In *Advances in Computers*, vol. 137, pp. 37-90. Elsevier, (2025).

B2. **Saifuzzaman, Munshi**, and Tajkia Nuri Ananna. '*Toward smart healthcare: challenges and opportunities in IoT and ML.*' In *IoT and ML for information management: A smart healthcare perspective*, pp. 325-355. Singapore: Springer Nature Singapore, (2024).

B3. Ananna, Tajkia Nuri, and **Munshi Saifuzzaman**. '*Introduction to Internet of Things.*' In *IoT and ML for Information Management: A Smart Healthcare Perspective*, pp. 1-49. Singapore: Springer Nature Singapore, (2024).

B4. Ananna, Tajkia Nuri; **Saifuzzaman, Munshi**; Chowdhury, Mohammad Jabed Morshed; Ferdous, Md Sadek: '*Managing Health Insurance Using Blockchain Technology*' In Healthcare Technologies, 2022), 'Blockchain Technology in e-Healthcare Management', Chap. 4, pp. 89-125.

## INDUSTRY EXPERIENCE

### ♣ Integrated Educational Information Management System [Web] *Jul 23 - Dec 23*

**Description:** Collaborated with clients to comprehend business requirements and contributed to the development team in successfully implementing a comprehensive system valued at 6 Crore BDT (USD 6M) for secondary and higher secondary education, catering to over 20 million students.

- ♠ Key contribution: Developed an examiner assignment system which can assign examiners for major public exams in Bangladesh, [**Teaching System**]
- ♠ Technologies: Spring Boot, Next.JS and TailwindCSS.
- ♠ Designed the documentation using entity relationship diagrams (ERDs), class diagrams, state diagrams, and data flow diagrams (DFDs).

## TEACHING EXPERIENCE (F = "Fall", S = "Spring")

♣ CS 2610: Developing Dynamic, Database-Driven, Web Applications — F24, S25, F25, S26
*TA, Utah State University*

- ♠ Covered TA duties including grading for over 180 students, bridging industry and academia.

♣ CSE 124: Business Application Design — F22
*Instructor, Daffodil International University*

- ♠ Designed, instructed, counseled and graded over 120 students where I taught students to design business applications with a focus on user psychology.

♣ CSE 216: Software Project II (Web Technology) — F22
*Instructor, Daffodil International University*

- ♠ Designed, instructed, counseled and graded over 120 students where I guided students through the end-to-end development of web applications using modern technologies.

## TECHNICAL SKILLS

♣ **Programming Language:** Java, Python 3, C, C++
♣ **Scripting:** LaTeX, HTML, CSS, JavaScript
♣ **Others:** Version Control (Git), Object Oriented Programming, Linux, Windows, SQL

♣ **Presentation:** Beamer, MS PowerPoint
♣ **Frameworks & Tools:** Android, TailwindCSS, Jupyter, Docker, Next.js, Spring Boot, MS Excel, Pandas, Firebase, MSSQL Server, DBeaver, Azure Data Studio, Postman, Streamlit, Django

## LEADERSHIP SKILLS

♣ **Class Representative** — Feb 17 – Aug 21
Served as a bridge between students and instructors for four consecutive years, facilitating communication and academic coordination by relaying announcements, decisions, concerns, and notices between students and instructors.

♣ **General Member** — Feb 17 – Aug 21
Volunteered as a member of the CSE Society at SUST, where I led and coordinated volunteers for five university fairs, ensuring smooth execution of society-led events.

## ACADEMIC PROJECTS

♣ **Fury in Motion |** *Python Streamlit* — Spring 25
An interactive platform for visualizing tornado patterns across the US, allowing users to filter by intensity, path, weather data, and more. [Project Link, Demo Link]

♣ **Digital Laundry |** *Java, Android, Firebase* — Summer 18, Fall 18
Digitized the laundry process by developing a system for doorstep laundry service management. [Project Link]

♣ **ChitChat P2P Messaging App |** *Java, Android* — Fall 24

A network-layer messaging app for Android that supports peer-to-peer (P2P) data exchange using Java Sockets. [Project Link, Demo Link]

♣ **Question Answering App |** *Java, Android, Firebase* Spring 20
Anonymous system of collecting data from people which is able to collect two types of responses: multiple-choice and short written answers. [Project Link]

♣ **SUST Online Examination System Frontend |** *React, Bootstrap* Fall 20, Spring 21
Developed the frontend of an online examination platform for my university, used for MCQ-based tests and assignment submissions. [Project Link]

♣ **AI Player |** *Java, MySQL, Android* Spring 20
A voice-controlled music player with an interactive UI; enables play/pause and track switching using voice or vibration controls. [Project Link]

♣ **CV Generator |** *Java Swing, SQLite* Spring 18
Generated customized CVs in PDF format based on user-provided data.

## LICENSE and CERTIFICATION

♣ **Coursera**
**Details:** Machine Learning [**Credential Id: GJM4HBC5DYWF**], Introduction to Git and GitHub [**Credential Id: 28DUAWHTMU6G**].

♣ **Cisco Networking Academy**
**Details:** Introduction to Cybersecurity [**Credential Id: 1009771690**].

## REFERENCE(s)

♣ **Prof. Tian Xie**
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♣ **Dr. Md. Jabed Morshed Chowdhury**
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